Agremind AI Models

Novel Smart Aquaculture and Agriculture System based on new AI Techniques

Agremind is an intelligent AI-powered platform designed to enhance aquaculture and agriculture productivity using advanced Machine Learning, Deep Learning, Time-Series Forecasting, Computer Vision, Optimization, and Large Language Models (LLMs).

The system is structured into domain-based intelligent modules, where each domain contains specialized AI models designed to solve real-world farming and aquaculture challenges.

The platform integrates:

- Computer Vision for disease detection and monitoring
- Time-Series Forecasting for growth and environmental prediction
- Machine Learning for yield, price, and resource optimization
- Reinforcement Learning for feeding and scheduling strategies
- LLM-based advisory system for intelligent decision support

This document presents the technical architecture of the AI models categorized by domain

1. Aquaculture Intelligence Module

The Aquaculture Intelligence Module focuses on fish and shrimp farming optimization, disease management, growth forecasting, water monitoring, feeding optimization, stock estimation, and economic forecasting.

Fish & Shrimp Disease Detection

Model 1: CNN-Based Classification

- **Type:** Deep Learning – Image Classification
- **Input:** Fish/Shrimp RGB image
- **Output:** Disease Class + Confidence Score
- **Purpose:** Early disease detection from pond images

Model 2: Hybrid CNN + Attention

- **Type:** Hybrid Deep Learning
- **Input:** Fish image
- **Output:** Disease Class + Confidence
- **Purpose:** Improve accuracy on fine-grained disease patterns

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Disease Severity Assessment

Model 3: Regression-Based Severity Estimation

- **Type:** CNN + Regression Head
- **Input:** Fish image
- **Output:** Severity Percentage (%)
- **Purpose:** Quantify disease progression level

Model 4: Segmentation-Based Severity

- **Type:** U-Net / DeepLabV3+
- **Input:** Fish image
- **Output:** Segmented infected area + Severity Level
- **Purpose:** Measure infected region for treatment planning

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Fish Growth Prediction

Model 5: Classical Time-Series

- **Type:** ARIMA / Prophet
- **Input:** Historical weight data
- **Output:** Future Growth Forecast
- **Purpose:** Short-term growth estimation

Model 6: Deep Learning Growth Model

- **Type:** LSTM / Transformer Time-Series
- **Input:** Growth history + Water parameters
- **Output:** Predicted biomass / weight
- **Purpose:** Long-term growth planning

Feeding Recommendation System

Model 7: ML-Based Feed Optimization

- **Type:** XGBoost / Random Forest Regression
- **Input:** Fish weight + Water conditions
- **Output:** Feed Quantity Recommendation
- **Purpose:** Optimize feed usage

Model 8: Reinforcement Learning Feeding Agent

- **Type:** DQN / PPO
- **Input:** Pond state (weight, temperature, oxygen)
- **Output:** Feeding Action Decision
- **Purpose:** Adaptive feeding strategy

Fish Counting (Stock Estimation)

Model 9: YOLO-Based Detection

- **Type:** Object Detection
- **Input:** Pond Image / Video
- **Output:** Fish Count
- **Purpose:** Stock estimation automation

Model 10: Transformer-Based Detection

- **Type:** DETR / RT-DETR
- **Input:** Underwater video frames
- **Output:** Count + Bounding Boxes
- **Purpose:** High-accuracy stock tracking

Water Quality Monitoring & Prediction

Model 11: Time-Series Water Forecasting

- **Type:** LSTM / GRU / TFT
- **Input:** pH, DO, Ammonia, Temperature (historical)
- **Output:** Future Water Quality Metrics
- **Purpose:** Prevent environmental stress

Model 12: Water Anomaly Detection

- **Type:** Isolation Forest / LSTM Autoencoder
- **Input:** Sensor Data Stream
- **Output:** Anomaly Flag + Risk Score
- **Purpose:** Early warning system

Harvest Time Prediction

Model 13: Harvest Forecast Model

- **Type:** XGBoost / LSTM
- **Input:** Growth + Market Price + Weather
- **Output:** Optimal Harvest Date
- **Purpose:** Maximize profit

Market & Production Forecasting

Model 14: Price Prediction

- **Type:** Transformer / LSTM
- **Input:** Historical Market Prices
- **Output:** Future Price Forecast
- **Purpose:** Selling strategy planning

Model 15: Production Yield Estimation

- **Type:** Random Forest / Neural Regression
- **Input:** Growth + Water + Feed
- **Output:** Expected Yield
- **Purpose:** Revenue estimation

Mortality & Risk Detection

Model 16: Behavioral Anomaly Detection

- **Type:** Autoencoder / One-Class SVM
- **Input:** Movement patterns + Sensor data
- **Output:** Mortality Risk Score
- **Purpose:** Early mortality prevention

Feed Conversion & Efficiency Analysis

Model 17: Feed Conversion Ratio (FCR) Predictor

- **Type:** Regression (XGBoost / RF)
- **Input:** Feed amount + Biomass gain + Water quality
- **Output:** Predicted FCR
- **Purpose:** Optimize feed efficiency and reduce cost

Oxygen Demand Forecasting

Model 18: Dissolved Oxygen Forecast Model

- **Type:** LSTM / Temporal Fusion Transformer
- **Input:** Historical DO + Temperature + Biomass
- **Output:** Future DO Levels
- **Purpose:** Prevent oxygen depletion events

Ammonia & Toxic Risk Prediction

Model 19: Ammonia Risk Estimator

- **Type:** Regression + Risk Classification
- **Input:** NH3, pH, Temperature, Feed rate
- **Output:** Toxicity Risk Level
- **Purpose:** Prevent chemical stress

Disease Spread Simulation

Model 20: Epidemiological Simulation Model

- **Type:** Compartmental Model (SIR-based) / Agent-Based Simulation
- **Input:** Infection rate + Density + Water conditions
- **Output:** Disease spread forecast
- **Purpose:** Containment planning

Smart Aeration Control

Model 21: Aerator Optimization Model

- **Type:** Reinforcement Learning (PPO / DQN)
- **Input:** DO level + Biomass + Weather
- **Output:** Aerator ON/OFF Schedule
- **Purpose:** Save energy while maintaining oxygen stability

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Biomass Estimation

Model 22: Biomass Estimation from Vision

- **Type:** CNN + Depth Estimation
- **Input:** Underwater image/video
- **Output:** Estimated Biomass (kg)
- **Purpose:** Non-invasive weight estimation

Water Pollution Visual Detection

Model 23: Pollution Detection Model

- **Type:** EfficientNet / ViT / Segmentation
- **Input:** Water surface image
- **Output:** Pollution Class + Area
- **Purpose:** Detect algae bloom / contamination

Stock Density Optimization

Model 24: Optimal Density Predictor

- **Type:** Regression + Constraint Optimization
- **Input:** Pond size + Water parameters + Species
- **Output:** Recommended stocking density
- **Purpose:** Prevent overcrowding

Economic Risk Modeling

Model 25: Profit & Risk Simulator

- **Type:** Monte Carlo Simulation + Regression
- **Input:** Yield forecast + Market price + Cost
- **Output:** Risk-adjusted profit range
- **Purpose:** Investment planning

Climate Stress Impact Model

Model 26: Heat Stress Index Model

- **Type:** XGBoost / LSTM
- **Input:** Water temperature + Humidity + Solar radiation
- **Output:** Stress score
- **Purpose:** Detect thermal risk

Sensor Drift Detection

Model 27: Sensor Reliability Model

- **Type:** Isolation Forest / Statistical Drift Detection
- **Input:** Long-term sensor readings
- **Output:** Drift flag
- **Purpose:** Detect faulty sensors

Multi-Parameter Pond Health Index

Model 28: Pond Health Scoring Model

- **Type:** Weighted ML Scoring / Ensemble
- **Input:** DO + NH3 + Temp + Biomass + Disease rate
- **Output:** Health Score (0–100)
- **Purpose:** Unified pond condition indicator

Feed Cost Optimization Under Market Fluctuation

Model 29: Cost-Aware Feed Optimization

- **Type:** Multi-objective Optimization (Genetic Algorithm / NSGA-II)
- **Input:** Feed price + Growth model + Market price
- **Output:** Optimal feed strategy
- **Purpose:** Maximize ROI

Survival Rate Prediction

Model 30: Survival Probability Model

- **Type:** Survival Analysis / Cox Regression
- **Input:** Historical mortality + Water parameters
- **Output:** Survival Probability Curve
- **Purpose:** Long-term production planning

2. Agriculture Intelligence Module

The Agriculture Intelligence Module focuses on crop monitoring, yield prediction, irrigation optimization, soil analysis, disease detection, economic forecasting, and farm-level decision support.

Crop Yield Prediction

Model 1: Machine Learning Yield Model

- **Type:** Random Forest / XGBoost / LightGBM
- **Input:** Soil (NPK, pH), Weather, Fertilizer, Irrigation
- **Output:** Predicted Yield (tons/hectare)
- **Purpose:** Estimate seasonal production

Model 2: Deep Learning Yield Model

- **Type:** LSTM / Transformer / DeepYield
- **Input:** Historical yield + Weather time-series
- **Output:** Yield Forecast
- **Purpose:** Long-term yield forecasting

Crop Price Forecasting

Model 3: Market Price Prediction

- **Type:** TFT / Informer / LSTM
- **Input:** Historical prices + Seasonality + Market indicators
- **Output:** Future price forecast
- **Purpose:** Selling strategy optimization

Weather & Resource Impact Modeling

Model 4: Yield Impact Analyzer

- **Type:** XGBoost / SVR
- **Input:** Rainfall, Temperature, Soil moisture, Fertilizer
- **Output:** Yield impact score
- **Purpose:** Identify limiting factors

Crop Disease Detection

Model 5: Leaf Disease Classification

- **Type:** CNN (ResNet / EfficientNet)
- **Input:** Leaf Image
- **Output:** Disease Class + Confidence
- **Purpose:** Early detection

Model 6: Field-Level Disease Detection

- **Type:** YOLO / Vision Transformer
- **Input:** Drone / Field images
- **Output:** Bounding boxes + Disease type
- **Purpose:** Large-scale monitoring

Model 7: Disease Segmentation

- **Type:** U-Net / SegFormer
- **Input:** Plant image
- **Output:** Infected area mask
- **Purpose:** Severity estimation

Weed Detection

Model 8: Weed Identification Model

- **Type:** YOLO / RetinaNet
- **Input:** Field images
- **Output:** Weed locations
- **Purpose:** Precision spraying

Ripeness Detection

Model 9: Harvest Readiness Model

- **Type:** CNN / YOLO
- **Input:** Fruit image
- **Output:** Ripeness class
- **Purpose:** Optimize harvest timing

Irrigation Management

Model 10: Irrigation Recommendation

- **Type:** Random Forest / SVR
- **Input:** Soil moisture + Weather + Crop stage
- **Output:** Water amount recommendation
- **Purpose:** Water optimization

Model 11: Smart Irrigation Forecast

- **Type:** LSTM / GRU
- **Input:** Soil moisture time-series
- **Output:** Future water requirement
- **Purpose:** Prevent stress

Soil Intelligence

Model 12: Soil Quality Classification

- **Type:** Random Forest / XGBoost
- **Input:** Nutrients + pH + Texture
- **Output:** Soil grade
- **Purpose:** Fertility evaluation

Model 13: Nutrient Deficiency Detection

- **Type:** CNN + Rule-based
- **Input:** Leaf image
- **Output:** Deficiency type (N/P/K)
- **Purpose:** Targeted fertilization

Field Zoning & Precision Farming

Model 14: Field Clustering

- **Type:** K-Means / Hierarchical Clustering
- **Input:** NDVI + Soil data + Yield history
- **Output:** Zone map
- **Purpose:** Precision agriculture

Fertilizer Optimization

Model 15: Fertilizer Recommendation

- **Type:** XGBoost / RF
- **Input:** Soil + Crop + Predicted yield
- **Output:** Fertilizer plan
- **Purpose:** Cost reduction

Model 16: Zone-Level Fertilizer Optimization

- **Type:** Multi-objective Optimization
- **Input:** Zone data + Budget
- **Output:** Optimized fertilizer allocation
- **Purpose:** Sustainable farming

Vegetation Index & Growth Monitoring

Model 17: NDVI Trend Forecast

- **Type:** LSTM / TFT
- **Input:** Time-stamped NDVI
- **Output:** Growth curve forecast
- **Purpose:** Early anomaly detection

Model 18: Growth Stress Detection

- **Type:** Isolation Forest / Autoencoder
- **Input:** Vegetation index series
- **Output:** Stress alert
- **Purpose:** Prevent yield loss

Harvest Planning

Model 19: Optimal Harvest Time

- **Type:** XGBoost / LSTM
- **Input:** Growth stage + Weather + Price
- **Output:** Recommended harvest date
- **Purpose:** Profit maximization

Production & Profit Modeling

Model 20: Production Estimator

- **Type:** Regression / Neural Network
- **Input:** Yield prediction + Field size
- **Output:** Total production estimate
- **Purpose:** Supply planning

Model 21: Profitability Predictor

- **Type:** Random Forest / XGBoost
- **Input:** Yield + Market price + Costs
- **Output:** Net profit estimate
- **Purpose:** Crop selection

Model 22: Risk-Adjusted Profit Model

- **Type:** Monte Carlo Simulation
- **Input:** Price volatility + Yield uncertainty
- **Output:** Profit distribution range
- **Purpose:** Investment decision

Climate & Stress Modeling

Model 23: Drought Risk Prediction

- **Type:** LSTM / RF
- **Input:** Rainfall deficit + Temperature
- **Output:** Drought risk score
- **Purpose:** Prevent crop stress

Model 24: Heat Stress Index

- **Type:** Regression / ML
- **Input:** Temperature + Humidity
- **Output:** Heat stress level
- **Purpose:** Protect sensitive crops

Crop Recommendation System

Model 25: Crop Suitability Model

- **Type:** Random Forest / XGBoost
- **Input:** Soil + Climate
- **Output:** Recommended crop list
- **Purpose:** Maximize yield potential

Supply Chain Forecasting

Model 26: Production-to-Market Forecast

- **Type:** Transformer Time-Series
- **Input:** Production forecast + Market demand
- **Output:** Supply forecast
- **Purpose:** Reduce waste

Carbon & Sustainability Monitoring

Model 27: Carbon Emission Estimator

- **Type:** Regression
- **Input:** Fertilizer usage + Fuel consumption
- **Output:** Estimated carbon footprint
- **Purpose:** Sustainable reporting

Pest Population Forecasting

Model 28: Pest Outbreak Prediction

- **Type:** LSTM / XGBoost
- **Input:** Weather + Historical pest data
- **Output:** Outbreak probability
- **Purpose:** Prevent infestation

Multi-Crop Optimization

Model 29: Crop Allocation Optimization

- **Type:** Linear Programming / Genetic Algorithm
- **Input:** Land area + Profit forecast
- **Output:** Optimal crop distribution
- **Purpose:** Farm-level optimization

Farm-Level Health Index

Model 30: Agriculture Health Score

- **Type:** Ensemble Scoring Model
- **Input:** Soil + Weather + Yield + Disease rate
- **Output:** Health Score (0–100)
- **Purpose:** Unified farm monitoring

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3. Weather & Environmental Intelligence Module

The Weather & Environmental Intelligence Module focuses on climate forecasting, environmental risk modeling, anomaly detection, resource stress prediction, and sustainability analytics for agriculture and aquaculture systems.

Weather Forecasting

Model 1: Short-Term Weather Prediction

- **Type:** LSTM / GRU
- **Input:** Historical temperature, humidity, rainfall
- **Output:** 1–7 day forecast
- **Purpose:** Operational planning

Model 2: Long-Range Climate Forecast

- **Type:** Transformer (Informer / TFT)
- **Input:** Multi-year climate data
- **Output:** Monthly seasonal forecast
- **Purpose:** Strategic planning

Rainfall Prediction

Model 3: Rainfall Intensity Model

- **Type:** XGBoost / LSTM
- **Input:** Pressure, humidity, temperature
- **Output:** Rainfall amount (mm)
- **Purpose:** Irrigation planning

Model 4: Extreme Rain Event Detection

- **Type:** Classification (RF / Neural Network)
- **Input:** Atmospheric indicators
- **Output:** Heavy rain probability
- **Purpose:** Flood prevention

Drought Monitoring

Model 5: Drought Risk Index

- **Type:** Regression / ML
- **Input:** Rainfall deficit + Soil moisture
- **Output:** Drought severity score
- **Purpose:** Early stress detection

Model 6: SPI-Based Drought Model

- **Type:** Statistical + ML Hybrid
- **Input:** Historical precipitation
- **Output:** Standardized Precipitation Index (SPI)
- **Purpose:** Long-term drought analysis

Temperature & Heat Stress

Model 7: Heat Wave Prediction

- **Type:** LSTM / Transformer
- **Input:** Temperature time-series
- **Output:** Heatwave probability
- **Purpose:** Crop & fish protection

Model 8: Heat Stress Index

- **Type:** Regression Model
- **Input:** Temperature + Humidity
- **Output:** Stress index value
- **Purpose:** Environmental impact scoring

Wind & Storm Modeling

Model 9: Wind Speed Forecast

- **Type:** GRU / ARIMA
- **Input:** Historical wind data
- **Output:** Wind forecast
- **Purpose:** Infrastructure safety

Model 10: Storm Risk Classification

- **Type:** Classification (XGBoost)
- **Input:** Pressure + Wind + Rain patterns
- **Output:** Storm risk level
- **Purpose:** Damage prevention

Flood Risk Modeling

Model 11: Flood Probability Model

- **Type:** Random Forest / LSTM
- **Input:** Rainfall + Terrain + River levels
- **Output:** Flood risk score
- **Purpose:** Disaster mitigation

Model 12: Runoff Simulation Model

- **Type:** Hydrological Simulation + ML
- **Input:** Soil type + Rainfall
- **Output:** Surface runoff estimation
- **Purpose:** Water management

Environmental Anomaly Detection

Model 13: Sensor Anomaly Detection

- **Type:** Isolation Forest
- **Input:** Multi-sensor environmental stream
- **Output:** Anomaly flag
- **Purpose:** Detect abnormal readings

Model 14: LSTM Autoencoder Drift Detection

- **Type:** Deep Learning Autoencoder
- **Input:** Time-series environmental data
- **Output:** Reconstruction error score
- **Purpose:** Detect abnormal environmental shifts

Soil & Moisture Forecasting

Model 15: Soil Moisture Forecast

- **Type:** LSTM / TFT
- **Input:** Historical soil moisture + rainfall
- **Output:** Future soil moisture levels
- **Purpose:** Irrigation optimization

Model 16: Evapotranspiration Prediction

- **Type:** Regression / ML
- **Input:** Temperature + Solar radiation + Wind
- **Output:** ET rate
- **Purpose:** Water loss estimation

Climate Impact Modeling

Model 17: Climate Impact on Yield

- **Type:** XGBoost / Neural Network
- **Input:** Climate forecast + Crop type
- **Output:** Yield impact estimate
- **Purpose:** Risk mitigation

Model 18: Climate Impact on Fish Health

- **Type:** LSTM / Regression
- **Input:** Water temperature + Climate trend
- **Output:** Stress risk score
- **Purpose:** Aquaculture protection

Air Quality Monitoring

Model 19: Air Quality Index Prediction

- **Type:** Regression / Time-Series
- **Input:** CO_2, NO_2, $PM_{2.5}$ levels
- **Output:** AQI value
- **Purpose:** Environmental monitoring

Model 20: Pollution Source Classification

- **Type:** Classification (RF / Neural Net)
- **Input:** Emission patterns
- **Output:** Pollution source category
- **Purpose:** Environmental control

Resource Stress Forecasting

Model 21: Water Scarcity Prediction

- **Type:** LSTM / XGBoost
- **Input:** Water availability + Demand
- **Output:** Scarcity risk score
- **Purpose:** Resource planning

Model 22: Resource Utilization Efficiency Model

- **Type:** Regression
- **Input:** Water + Energy usage + Yield
- **Output:** Efficiency score
- **Purpose:** Sustainability optimization

Carbon & Sustainability Analytics

Model 23: Carbon Emission Forecast

- **Type:** Regression / Time-Series
- **Input:** Fuel + Fertilizer usage
- **Output:** CO_2 projection
- **Purpose:** Environmental reporting

Model 24: Climate Risk Composite Index

- **Type:** Ensemble Scoring Model
- **Input:** Drought + Flood + Heat + Storm indicators
- **Output:** Overall climate risk score
- **Purpose:** Decision dashboard

Vegetation & Environmental Monitoring

Model 25: NDVI Environmental Trend Model

- **Type:** LSTM / Transformer
- **Input:** Satellite vegetation index
- **Output:** Vegetation trend forecast
- **Purpose:** Ecosystem monitoring

Model 26: Land Degradation Detection

- **Type:** CNN / Segmentation
- **Input:** Satellite imagery
- **Output:** Degradation map
- **Purpose:** Long-term sustainability tracking

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4. Energy & Smart Resource Optimization Module

The Energy & Smart Resource Optimization Module focuses on intelligent energy forecasting, pump scheduling, renewable integration, cost minimization, and multi-objective optimization across agriculture and aquaculture systems.This module ensures sustainable operation while minimizing cost and maximizing productivity.

Energy Consumption Forecasting

Model 1: Farm Energy Consumption Prediction

- **Type:** LSTM / GRU
- **Input:** Historical energy usage + Operational logs
- **Output:** Future energy consumption (kWh)
- **Purpose:** Load planning

Model 2: Equipment-Level Energy Forecast

- **Type:** XGBoost / RF
- **Input:** Pump runtime + Aerator usage + Sensors
- **Output:** Device-level energy estimate
- **Purpose:** Device optimization

Smart Pump Scheduling

Model 3: Pump Operation Optimization

- **Type:** Reinforcement Learning (DQN / PPO)
- **Input:** Water level + Soil moisture + Energy price
- **Output:** Pump ON/OFF schedule
- **Purpose:** Reduce energy cost

Model 4: Irrigation-Energy Joint Optimization

- **Type:** Multi-objective Optimization (NSGA-II)
- **Input:** Water demand + Energy cost
- **Output:** Optimal irrigation-energy schedule
- **Purpose:** Cost-efficient water delivery

Solar Energy Integration

Model 5: Solar Power Generation Forecast

- **Type:** LSTM / TFT
- **Input:** Solar radiation + Weather forecast
- **Output:** Predicted solar output (kW)
- **Purpose:** Renewable planning

Model 6: Solar Panel Efficiency Model

- **Type:** Regression
- **Input:** Temperature + Radiation + Panel age
- **Output:** Efficiency percentage
- **Purpose:** Maintenance planning

Battery & Storage Optimization

Model 7: Battery Charge Forecast

- **Type:** Time-Series Model
- **Input:** Charge cycles + Load demand
- **Output:** Future battery level
- **Purpose:** Storage planning

Model 8: Battery Dispatch Optimization

- **Type:** Linear Programming
- **Input:** Solar forecast + Load + Grid price
- **Output:** Optimal charge/discharge schedule
- **Purpose:** Cost minimization

Energy Cost Modeling

Model 9: Dynamic Energy Pricing Forecast

- **Type:** LSTM / XGBoost
- **Input:** Historical tariff data
- **Output:** Future energy price
- **Purpose:** Smart scheduling

Model 10: Energy Cost Estimator

- **Type:** Regression
- **Input:** Predicted usage + Tariff
- **Output:** Cost projection
- **Purpose:** Budget planning

Aeration Energy Optimization (Aquaculture)

Model 11: Smart Aerator Control

- **Type:** Reinforcement Learning
- **Input:** DO levels + Biomass + Energy price
- **Output:** Aerator schedule
- **Purpose:** Maintain oxygen with minimal power

Water-Energy Nexus Modeling

Model 12: Water Pump Energy Demand Model

- **Type:** Regression / ML
- **Input:** Pump depth + Flow rate
- **Output:** Energy required (kWh)
- **Purpose:** Efficiency evaluation

Model 13: Water Distribution Optimization

- **Type:** Linear Programming
- **Input:** Field demand + Pump capacity
- **Output:** Distribution plan
- **Purpose:** Minimize pumping cost

Multi-Objective Farm Optimization

Model 14: Yield-Energy Tradeoff Model

- **Type:** Genetic Algorithm
- **Input:** Yield forecast + Energy usage
- **Output:** Optimal strategy set
- **Purpose:** Balance productivity and cost

Model 15: Profit-Energy Optimization

- **Type:** Monte Carlo + Optimization
- **Input:** Yield + Market price + Energy cost
- **Output:** Risk-adjusted optimal plan
- **Purpose:** Maximize ROI

Equipment Health & Energy Efficiency

Model 16: Motor Efficiency Detection

- **Type:** Anomaly Detection (Autoencoder)
- **Input:** Current + Voltage + Runtime
- **Output:** Efficiency anomaly score
- **Purpose:** Prevent energy waste

Model 17: Predictive Maintenance Model

- **Type:** LSTM / Classification
- **Input:** Vibration + Temperature + Usage
- **Output:** Failure probability
- **Purpose:** Reduce downtime

Carbon & Sustainability Optimization

Model 18: Energy Carbon Footprint Estimator

- **Type:** Regression
- **Input:** Energy consumption + Source mix
- **Output:** CO_2 emission estimate
- **Purpose:** Sustainability tracking

Model 19: Low-Carbon Optimization Model

- **Type:** Multi-objective Optimization
- **Input:** Energy mix + Operational demand
- **Output:** Low-emission plan
- **Purpose:** Environmental compliance

Grid Interaction Modeling

Model 20: Grid Load Prediction

- **Type:** LSTM
- **Input:** Historical farm demand
- **Output:** Grid demand forecast
- **Purpose:** Load balancing

Model 21: Peak Shaving Optimization

- **Type:** Linear Programming
- **Input:** Load profile + Tariff
- **Output:** Reduced peak demand plan
- **Purpose:** Reduce peak penalties

Smart Microgrid Management

Model 22: Microgrid Control Model

- **Type:** Reinforcement Learning
- **Input:** Solar + Battery + Grid + Load
- **Output:** Optimal energy flow
- **Purpose:** Autonomous energy management

Water-Solar Integrated Optimization

Model 23: Solar-Powered Irrigation Planner

- **Type:** Hybrid Optimization
- **Input:** Solar forecast + Irrigation demand
- **Output:** Solar-aligned irrigation schedule
- **Purpose:** Zero-grid irrigation

Energy Efficiency Scoring

Model 24: Farm Energy Efficiency Index

- **Type:** Ensemble Scoring Model
- **Input:** Consumption + Yield + Water use
- **Output:** Efficiency score (0–100)
- **Purpose:** Unified sustainability KPI

Scenario Simulation Engine

Model 25: Energy Scenario Simulator

- **Type:** Monte Carlo Simulation
- **Input:** Price volatility + Weather uncertainty
- **Output:** Cost distribution scenarios
- **Purpose:** Strategic decision support

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5. LLM & Cognitive Intelligence Layer

The LLM & Cognitive Intelligence Layer acts as the central reasoning engine of the Agremind platform.
It integrates all AI domain outputs and transforms them into actionable, explainable, and strategic decisions.

Model 1: Conversational & Advisory Intelligence

- **Purpose:** Provide real-time domain-aware advisory support for farmers and aquaculture operators.
- **Core Model:** Fine-Tuned LLM (LLaMA / Mistral / GPT)
- **Input:** User natural language query **,** Optional system data (yield, water, weather, energy)
- **Output:** Context-aware recommendation **,** Action steps **,** Clarified explanations
- **Capabilities:** Agriculture advisory **,** Aquaculture advisory , Energy & sustainability advice , Multilingual support

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Model 2: Retrieval-Augmented Knowledge System (RAG)

- **Purpose:** Ensure accurate, evidence-based responses using structured agricultural and aquaculture knowledge.
- **Core Model :** LLM + Embedding Model + Vector Database
- **Input:** User query
- **Output:** Retrieved relevant knowledge chunks **,** Grounded response with factual support
- **Capabilities:** Research-based answers **,** Disease treatment guidance , Fertilizer & feeding protocols , Policy and environmental guidelines

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Model 3: Explainable AI & Decision Reasoning Engine

- **Purpose :** Translate raw ML predictions into human-understandable decisions.
- **Core Model :** LLM + Model Metadata + SHAP/Feature Importance
- **Input:** Model prediction (yield, disease risk, anomaly, etc.) **,** Feature importance scores
- **Output:** Plain-language explanation , Cause analysis , Recommended action
- **Capabilities:** Alert explanation , Risk justification , Confidence-based advisory , Cross-domain reasoning

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- **Model 4: Strategic Planning & Multi-Domain Intelligence**
- **Purpose :** Provide high-level strategic recommendations using outputs from multiple AI modules.
- **Core Model :** LLM + Simulation Results + Forecast Models
- **Input :** Yield forecast , Market prediction , Climate risk , Energy cost
- **Output :** Integrated strategy , Risk-adjusted plan , Priority task list
- **Capabilities :** Seasonal planning , Harvest strategy , Cost optimization , Long-term sustainability planning

Models Distribution by Module

Module	Number	Main AI Types Used	Core Purpose
Aquaculture Intelligence	28	CNN, LSTM, XGBoost, Regression, Classification	Fish health monitoring, water quality prediction, feeding optimization
Agriculture Intelligence	30	CNN, Random Forest, Gradient Boosting, Time-Series	Crop disease detection, yield prediction, irrigation optimization
Weather & Environmental Intelligence	22	LSTM, ARIMA, Prophet, Regression	Weather forecasting, environmental risk detection
Energy & Smart Resource Optimization	18	Optimization Algorithms, ML Regression, Forecasting	Solar prediction, energy load forecasting, smart irrigation control
LLM & Cognitive Intelligence Layer	4	LLM, RAG, NLP, Prompt Engineering	Advisory system, reasoning, smart reporting
Total AI Models	**102 Models**	Hybrid AI Architecture	Fully Integrated Smart Agro-Aqua Platform

AI Architecture Layers

Layer	Description
Perception Layer	Vision-based models (CNN) for disease and anomaly detection
Prediction Layer	Time-series forecasting & regression models
Optimization Layer	Resource and energy optimization algorithms
Cognitive Layer	LLM-based reasoning and advisory system